

***EMOTION-BASED MOVIE
RECOMMENDATION SYSTEM***
BY TEAM 23

Problem Statement

- Users spend a lot of time searching for movies matching their mood.
- Existing recommendation systems mainly depend on search history and ratings.
- They do not understand the real-time emotions of users.
- This project detects facial emotions using a webcam.
- Based on detected emotion and language preference, suitable movies are recommended.
- The system also provides trailers, ratings, and OTT platform details for a better user experience.

Objectives

- To develop a web application for emotion-based movie recommendation.
- To detect user emotions using facial expressions.
- To recommend movies according to mood and language selection.
- To provide trailers, ratings, and OTT platform information.
- To create an interactive Netflix-style user interface.
- To combine AI, Computer Vision, and Web Technologies in one system.

Feasibility

Technical Feasibility

- Uses OpenCV, Deep Face, Flask, HTML, CSS, and JavaScript.
- Supports real-time emotion detection and movie recommendation.

Economic Feasibility

- Uses open-source tools and free APIs.
- Requires only a webcam and internet connection.

Operational Feasibility

- Easy to use and user-friendly interface.
- Fast and interactive recommendation process.

Tools and Technologies

Frontend

HTML — Structure of web pages

CSS — Netflix-style UI design

JavaScript — Webcam capture and dynamic content

Backend

Python — Core programming language

Flask — Backend server and routing

AI & Computer Vision

Deep Face — Facial emotion detection

OpenCV — Webcam and image processing

APIs Used

TMDB API — Movie details, trailers, ratings, OTT platforms

YouTube — Trailer playback

Development Tools

VS Code — Code editor

Libraries: Flask, OpenCV, DeepFace, NumPy, Requests

System Flow

Camera → Emotion Detection → Movie Recommendation → Trailer & OTT Display

Relevance of Project

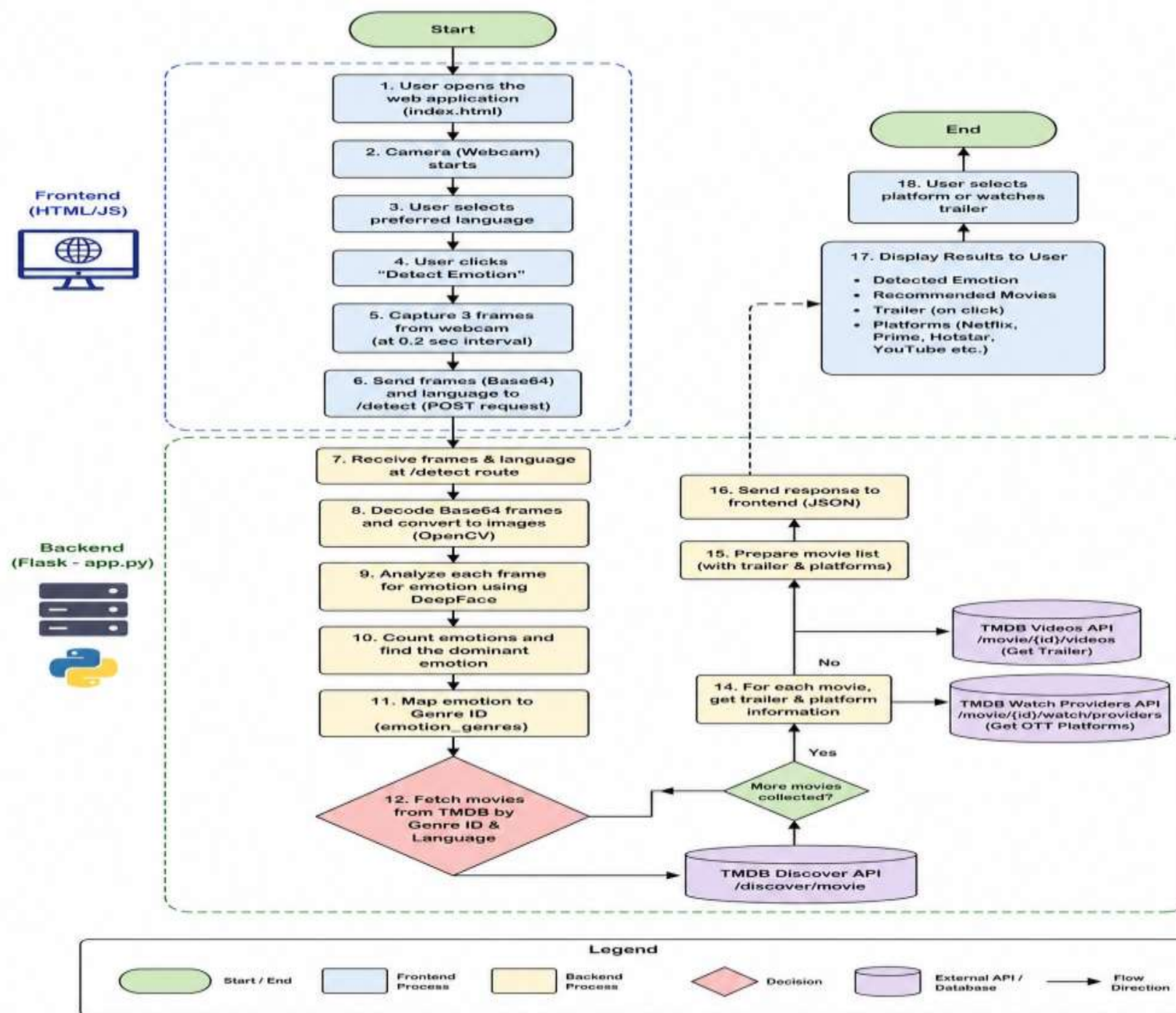
- Provides personalized movie recommendations based on emotions.
- Improves user entertainment experience.
- Combines AI, Computer Vision, and Web Development.
- Useful in modern OTT and streaming applications.
- Demonstrates real-world application of Artificial Intelligence.
- Can be extended for music or content recommendation systems.

Design

- Frontend designed using HTML, CSS, and JavaScript.
- Backend developed using Flask in Python.
- Webcam captures user facial expressions.
- Deep Face detects emotions from captured frames.
- TMDB APIs fetch movie details, trailers, and OTT platforms.
- Results displayed in Netflix-style poster format.

Implementation and demo

- User opens the web application.
- Webcam captures facial expressions.
- User selects preferred language.
- DeepFace analyzes emotions using captured frames.
- Emotion is mapped to movie genres.
- TMDB APIs fetch recommended movies.
- Posters, trailers, ratings, and OTT platforms are displayed.



Contribution

- Developed a real-time emotion-based recommendation system.
- Integrated AI with web application development.
- Provided personalized movie recommendations.
- Added multi-language support.
- Included trailer and OTT platform integration.
- Improved user interaction with modern UI design

Learning and Challenges

Learning Outcomes

- Learned Flask web development.
- Understood API integration and JSON handling.
- Worked with OpenCV and Deep Face.
- Improved frontend and backend integration skills.
- Gained knowledge in AI and recommendation systems.

Challenges Faced

- Handling real-time webcam processing.
- Improving emotion detection accuracy.
- Managing frontend-backend communication.
- Integrating multiple APIs together.
- Designing a responsive user interface.

Involvement in Project Work & Teamwork

Involvement in Project Work

- Contributed to project planning, design, and implementation.
- Participated in frontend and backend development tasks.
- Worked on emotion detection and movie recommendation features.
- Assisted in API integration and UI design.

Teamwork

- Collaborated effectively with team members during development.
- Shared responsibilities for coding, testing, and debugging.
- Coordinated ideas and improvements for better project performance.
- Maintained good communication and teamwork throughout the project.

Thank you!!!!